CHUKWUEBUKA BENJAMIN UGWU

Full-Stack & Applied AI Engineer | LLM Evaluation & RLHF Specialist (Python / FastAPI / React / Next.js)

Kaduna, Nigeria (UTC+1 / EMEA Overlap) | Ready for Global Contractor / Deel / EOR Onboarding

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PROFESSIONAL SUMMARY

Full-Stack & Applied AI Engineer and LLM Evaluation Specialist with a B.Tech in Computer Science and hands-on production experience architecting low-latency web applications, adaptive machine learning systems, and structured AI evaluation pipelines. Proven expertise in Reinforcement Learning from Human Feedback (RLHF), Direct Preference Optimization (DPO), Supervised Fine-Tuning (SFT) golden response authoring, Chain-of-Thought (CoT) verification, and hallucination/factuality auditing across STEM, coding. Adept at shipping end-to-end applications using Python (FastAPI, Flask), JavaScript/TypeScript (React, Next.js, Node.js), PostgreSQL, and Redis.

CORE COMPETENCIES & TECHNICAL SKILLS

AI Training, RLHF & LLM Evaluation: RLHF, Direct Preference Optimization (DPO), SFT Golden Response Authoring, Rubric Calibration & Inter-Rater Reliability (Cohen's Kappa), Chain-of-Thought (CoT) Reasoning Verification, Hallucination & Factuality Auditing, Red-Teaming, Multi-Turn Agent Trajectory Grading.

Languages : Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3, Bash | Native/Fluent

Backend & Distributed Systems: FastAPI, Flask, Node.js, RESTful API Architecture, Pydantic Schema Validation, Webhooks, Microservices, Stream Deduplication.

Frontend & Full-Stack Web: React.js, Next.js, Tailwind CSS, State Management, Responsive & Offline-First UI Design, Web3Forms API.

Databases, Caching & ML Engines: PostgreSQL (Normalized Schema Design, Indexing, Query Optimization), Redis (Low-Latency Session Caching), Bayesian Knowledge Tracing (BKT), Reinforcement Learning (RL), TensorFlow.

Testing, QA & DevOps Infrastructure: Pytest, Postman (API Contract Testing), End-to-End Functional & Negative Testing, Docker, WSL2, Git, GitHub Actions (CI/CD), Vercel, Custom DNS, iPerf3.

PROFESSIONAL EXPERIENCE

Lead Full-Stack & Applied AI Engineer | AI Adaptive Learning & Multilingual Evaluation Platform *(Dec 2025 – Oct 2026)*

- Architected and deployed an end-to-end multilingual AI adaptive learning system (supporting English, Hausa, Igbo, and Yoruba) covering 100-level introductory Python programming, STEM, History, and Literature.
- Engineered the core inference backend in Python (FastAPI/Flask) combining Bayesian Knowledge Tracing (BKT) for real-time mastery probability estimation with a Reinforcement Learning (RL) policy engine to sequence optimal learning trajectories.
- Authored, calibrated, and audited 500+ structured domain rubrics, golden reference solutions, and step-by-step Chain-of-Thought (CoT) explanations across programming and humanities modules to eliminate content hallucinations.
- Designed a normalized PostgreSQL relational schema for longitudinal learner telemetry and integrated Redis for low-latency session caching, maintaining p95 API latency under 120ms and improving student completion rates by 35%.

AI Quality Assurance & Technical Evaluation Specialist | Independent Contract / Remote Evaluation Pipelines *(Aug 2026 – Present)*

- Evaluated multi-turn AI coding agent trajectories, LLM outputs, and complex system design reasoning across Python, JavaScript, and SQL benchmarks using strict rubric scoring and Cohen's Kappa calibration standards.
- Conducted side-by-side RLHF/DPO preference ranking, identifying subtle logic bugs, edge-case regressions, security vulnerabilities, and instruction-following failures in model-generated code and analytical prose.
- Designed deterministic evaluation harnesses and end-to-end functional/negative QA test suites (verifying concurrency locks, prescription-only validation gates, and boundary conditions) across web and e-pharmacy workflows.

Forward-Deployed Full-Stack Developer | Bfarmcon (Commercial Agro-Supply Web Platform) *(Aug 2026 – present)*

- Designed, built, and shipped a production commercial web platform using React.js, Next.js, and JavaScript for an agricultural commodity enterprise.

- Engineered automated lead-capture and order-routing pipelines integrating Web3Forms API and WhatsApp Business routing with idempotency guards and UI debouncing, reducing failed or duplicate mobile submissions on 3G networks to 0%.
- Configured automated CI/CD deployment pipelines on Vercel, custom domain DNS routing, XML sitemaps, and Google Search Console indexing to maximize organic search visibility.

Systems Performance & Benchmarking Engineer | Network Throughput & Latency Benchmarking Suite *(Jul 2026 – Sep 2026)*

- Built an automated network throughput, memory, and latency benchmarking suite in Python leveraging iPerf3 and socket protocols across Windows 11 and WSL2 client-server environments.
- Automated multi-run stress testing, packet-loss detection, and statistical anomaly reporting, reconciling performance drift across constrained hardware profiles.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science | Federal University of Technology, Minna (FUTMinna), Nigeria *(Sep 2021 – Aug 2026)*

- Thesis: Development of an AI-Powered Adaptive Learning System for Personalized Education using Bayesian Knowledge Tracing and Reinforcement Learning.
- Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Database Management Systems, Computer System Performance & Benchmarking, Software Engineering.
- Certifications: Certified Frontend Web Developer – IRL IT Solutions (Nov 2022).